

Indian Institute of Technology Bombay

PH 544 - EndSem Exam
Marks 45

General relativity
LC-301

Date: 28/04/2026
Time: 8:00-11:00

1. For $R_{\alpha\beta\gamma\delta} = K(g_{\alpha\gamma}g_{\beta\delta} - g_{\alpha\delta}g_{\beta\gamma})$, where K is a scalar function. Show that K is constant. [3]

2. Given

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu,$$

where A^μ is a four vector. Is $F_{\mu\nu}$ a second rank tensor? Please justify your answer mathematically. [3]

3. A test particle of mass m is tied to a massless string held by an observer at infinity in Schwarzschild spacetime. The particle is at a fixed coordinate distance r_0 . Given the four-acceleration, $a^\mu \equiv u^\nu \nabla_\nu u^\mu$, compute the magnitude of the force the particle experiences. Is this the same as the force that the observer feels? Explain. [3]

4. The proper distance of a galaxy cluster XLSSC122 is 5.3 Gpc. Assuming the value of the Hubble constant at current epoch as $H_0 = 70 \text{ km/sec} * \text{Mpc}$, compute the relative velocity of this cluster from us. Please comment on your result. [3]

5. An observer falls radially inward toward a BH of mass M , starting with zero kinetic energy at infinity. How much time will it take, as measured by its own clock, to pass between the radii $6M$ and $2M$? Please calculate in Geometric units with $G=c=1$. [4]

6. Consider the following metric that describes a weak gravitational field:

$$ds^2 = \left(1 + \frac{2\phi}{c^2}\right) dt^2 - dx^2 - dy^2 - dz^2$$

where $\phi(x, y, z)/c^2 \ll 1$. Compute the Ricci tensor up to the lowest non-trivial order in ϕ/c^2 . [4]

7. A test particle orbiting around the BH will emit gravitational waves leading to the shrinkage of the circular orbit. To estimate the total energy radiated in the form of gravitational waves, you can consider the motion of test particle of mass m orbiting around a Schwarzschild BH with mass M . We assume the initial orbital radius in Schwarzschild coordinate to be $r_0 \gg M$ and the particle's orbit keeps shrinking until it plunges into the BH. Compute the total energy emitted via gravitational waves if the energy during the plunge phase is negligible. Clearly state your assumptions. [5]

8. (a) An object O moves at speed u' in frame S' . S' frame moves at speed v with respect to S in the same direction as O . Draw the spacetime diagram showing (ct, x) and (ct', x') axes. Using the space-time diagram, obtain the speed of the object (u) observed in frame S . [4]
- (b) A medium with refractive index n moves at v w.r.t the lab frame along x - x' axis. Light travels in this medium along x - x' direction. Using result from Part (a), calculate the speed of light u as measured by an observer in the lab frame. For low speeds ($v \ll c$), find the first-order approximation for u in terms of v , c , and n . Comment on the result. [2]

9. (a) Given the electromagnetic field tensor,

$$F^{\alpha\beta} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

Show that the Lorentz force in STR is given by $\alpha = i$ in the following expression

$$\frac{dp^\alpha}{d\tau} = qF^{\alpha\beta}u_\beta$$

where p^α and u^β are the four momentum and four velocity vectors respectively. [3]

- (b) Consider an infinite non-conducting rod of radius R with uniform volume charge density ρ . Compute the fields due this at $r > R$. Now consider a moving frame with velocity v with respect to the rod. The relative motion is along the length of the rod. Using (a), compute the field due to the charge distribution in this moving frame. [3]
10. The hyperbolic plane is defined by the metric

$$ds^2 = \frac{1}{y^2}(dx^2 + dy^2) \quad y \geq 0$$

is an example of the 2-D surface. [8]

- (a) Show that the points on the x -axis are an infinite distance from any point in the upper half-plane. [1]
- (b) Write down the geodesic equations. [3]
- (c) Obtain the shape of the geodesics in x - y plane. [4]

Useful mathematical relations

$$\begin{aligned}
 \frac{d^2 q^\nu}{d\lambda^2} + \Gamma_{\mu\kappa}^\nu \frac{dq^\mu}{d\lambda} \frac{dq^\kappa}{d\lambda} &= 0 \\
 \Gamma_{\gamma\epsilon}^\alpha &= \frac{1}{2} g^{\alpha\beta} (\partial_\gamma g_{\beta\epsilon} + \partial_\epsilon g_{\beta\gamma} - \partial_\beta g_{\gamma\epsilon}) \\
 \nabla_\alpha A_\beta &= \partial_\alpha A_\beta - \Gamma_{\alpha\beta}^\delta A_\delta = A_{\beta,\alpha} - \Gamma_{\alpha\beta}^\delta A_\delta \\
 \nabla_\alpha A^\beta &= A_{,\alpha}^\beta + \Gamma_{\alpha\delta}^\beta A^\delta \\
 \nabla_\alpha T_{\beta\gamma} &= T_{\beta\gamma,\alpha} - \Gamma_{\alpha\beta}^\delta T_{\delta\gamma} - \Gamma_{\alpha\gamma}^\delta T_{\beta\delta} \\
 R_{\beta\mu\nu}^\alpha &= \Gamma_{\beta\mu,\nu}^\alpha - \Gamma_{\beta\nu,\mu}^\alpha + \Gamma_{\sigma\nu}^\alpha \Gamma_{\beta\mu}^\sigma - \Gamma_{\sigma\mu}^\alpha \Gamma_{\beta\nu}^\sigma \\
 R_{\alpha\beta\gamma\delta} &= -R_{\beta\alpha\gamma\delta} = -R_{\alpha\beta\delta\gamma} = R_{\beta\alpha\delta\gamma} \\
 R_{\alpha\beta\gamma\delta} + R_{\alpha\gamma\delta\beta} + R_{\alpha\delta\beta\gamma} &= 0 \\
 \nabla_\epsilon R_{\beta\gamma\delta}^\alpha + \nabla_\gamma R_{\beta\delta\epsilon}^\alpha + \nabla_\delta R_{\beta\epsilon\gamma}^\alpha &= 0 \\
 R_{\beta\alpha\mu}^\alpha = R_{\beta\mu} & \quad R_{\beta\mu} = \Gamma_{\beta\mu,\alpha}^\alpha - \Gamma_{\beta\alpha,\mu}^\alpha + \Gamma_{\beta\mu}^\sigma \Gamma_{\sigma\alpha}^\alpha - \Gamma_{\beta\alpha}^\sigma \Gamma_{\mu\sigma}^\alpha \\
 R_{\alpha\beta\gamma\delta} &= \frac{1}{2} (g_{\beta\gamma,\alpha\delta} + g_{\alpha\delta,\beta\gamma} - g_{\beta\delta,\alpha\gamma} - g_{\alpha\gamma,\beta\delta}) + g_{\mu\nu} \Gamma_{\alpha\gamma}^\nu \Gamma_{\beta\delta}^\mu - g_{\mu\nu} \Gamma_{\alpha\delta}^\nu \Gamma_{\beta\gamma}^\mu \\
 ds^2 &= \left(1 - \frac{2GM}{r}\right) dt^2 - \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 - r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \\
 \frac{\tilde{E}^2}{c^2} &= \left(\frac{dr}{d\tau}\right)^2 + c^2 \left(1 - \frac{2GM}{c^2 r}\right) \left(1 + \frac{\ell^2}{r^2 c^2}\right) \tag{1}
 \end{aligned}$$

All the best